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A Classification Method for Cloud-Contaminated Time-Series Remote Sensing Imagery

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***Abstract*—This paper proposes a joint cloud removal and land cover classification method for time-series remote sensing images to address classification challenges caused by cloud contamination. The method features a dual-modal algorithm for cloud restoration: 1) a superpixel-based regression model for local reconstruction of overlapping cloud areas , and 2) an isophote network to ensure smooth restoration of non-overlapping areas. For classification, the framework fuses multi-modal features (color, texture, statistics) into a descriptor and uses a stratified Bootstrap random forest to mitigate overfitting and address the problem of class sample imbalance. Experiments on a Landsat 8 dataset show the classification difference rate of reconstructed images is under 3.05% for complex terrain and 0.87% for homogeneous terrain. The method demonstrates improved detail preservation over classifiers like SVM and KNN , offering a viable path for dynamic land cover monitoring in frequently cloudy regions.**

Keywords—time-series image reconstruction; isophote network; multi-modal features; Bootstrap ensemble learning; land cover classification.

I. Introduction

Satellite imagery is indispensable for applications in agriculture, forestry, urban planning, and disaster monitoring [1]. However, cloud cover inherently constrains the quality of optical remote sensing data by causing data gaps and radiometric inconsistencies , which poses significant challenges for time-sensitive applications.

Currently, two main approaches exist for classifying cloud-contaminated images: direct classification and classification after cloud removal. Direct classification, an end-to-end paradigm, uses machine learning models to bypass the removal step. While this reduces the risk of semantic information loss , it requires large, costly labeled datasets and can be misled by the optical distortions caused by clouds , especially when fusing multi-modal data [2].

The more traditional approach, classification after cloud removal [3], first repairs contaminated areas and then classifies the restored image. This method's limitation is its heavy dependence on high-quality, cloud-free reference images, which are often unavailable in frequently cloudy regions[4]. Furthermore, overlapping cloud cover between target and reference images can interrupt or introduce errors into the reconstruction process.

To address the aforementioned challenges, this paper proposes a framework for classifying cloudy time-series remote sensing images that integrates dual-path cloud removal with multi-modal classification:

1） Adaptive Reference Image Selection: Determines repairability through temporal logical operations on cloud masks and implements a middle-frame priority strategy to enhance the effectiveness of the reference image.

2） Context-Aware Reconstruction: A regression model based on superpixel segmentation uses color structure constraints to establish a local mapping, resolving the issue of cloud area overlap. An isophote network constraint performs convolution over the entire image to create a system of linear equations, enabling large-area cloud removal while maintaining spatial continuity in non-overlapping areas, provided a cloud-free reference image is available.

3） Robust Classification Mechanism: Fuses color, texture, and multi-scale statistical features to create a 20-dimensional descriptor, which is combined with a stratified Bootstrap integrated random forest to suppress class imbalance.

II. Method

As shown in Figure 1, the algorithm in this paper can be divided into two stages: the cloud-contamination repair stage and the land cover type classification stage.

For a series of cloudy time-series remote sensing images to be classified, the cloud masks are first used to determine if cloud removal can be completed and to understand the relationships between the masks. If possible, cloud removal begins on a priority basis. When removing clouds from the i-th image, the (i-1)-th and (i+1)-th images are selected as its reference images, and different cloud removal algorithms are chosen based on the relationship between their cloud masks. A classifier is trained using cloud-free sample points, multi-scale features are extracted, and a training set is constructed using bootstrap resampling for classifier training.

A. Cloud-Contaminated Image Repair

Cloud removal feasibility for the time-series image set is determined by performing a logical AND operation on their cloud masks. Images are processed in chronological order, prioritizing middle images (e.g., 2, 3, 4). The reference image for reconstruction is selected based on the AND operation results, prioritizing images with overlapping cloud regions. Figure 2 shows the cloud removal process that combines complementary two methods.

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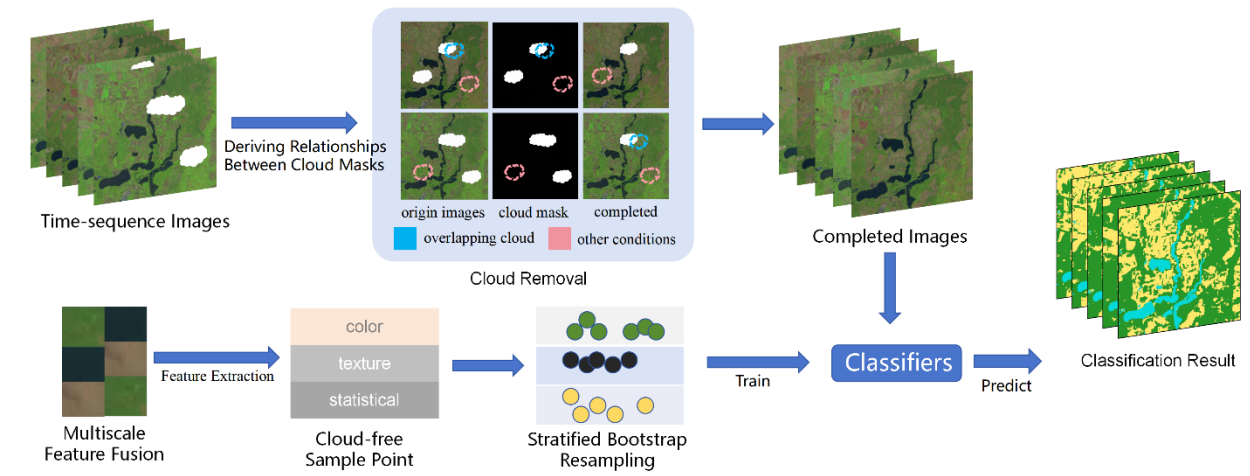


Fig. 1. Flow chart of the classification method for cloud-contaminated time-series remote sensing imagery.

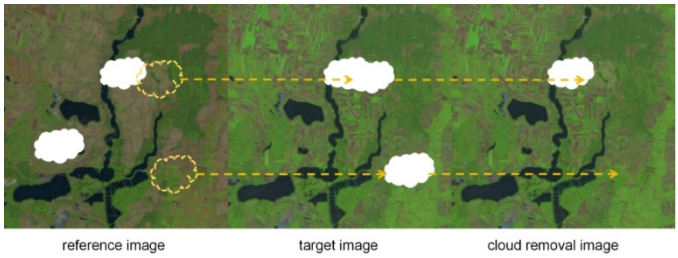


Fig. 2. The application of different algorithms: The clouds in the upper part of this simulated cloud map overlap, while those in the lower part do not. Verify the two cloud removal methods respectively.

For overlapping cloud regions: as in figure 3, a superpixel segmentation-based regression model [5] (in our previous papers, using an improved density-feat SLIC algorithm) fills target image regions (I^A_{target}, I^B_{target}, I^C_{target}) using cloud-free pixels from corresponding segments in the reference image (J^A_{known} ,J^B_{known} ,J^C_{known}). Regression models are built per superpixel segment using known pixels (I^X_{known}) and solved to predict unknown pixels (I^X_{target}) from the reference (J^X_{target}).This method handles complex clouds but may result in faint outlines and poorer spatial/textural continuity.

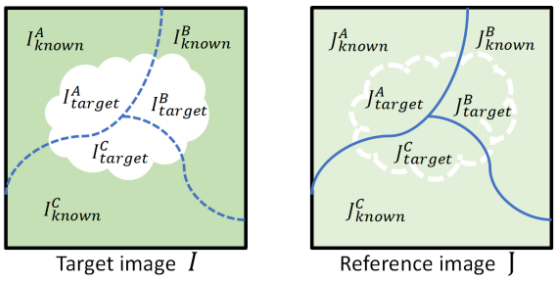


Fig. 3. Regression fitting analysis based on color structure constraints

For non-overlapping regions: A isophote-based cloud removal method [6] is used. this method has superior spatial continuity and smooth transitions but requires cloud-free reference image regions. The reference region's isophote network ($\nabla J^{\perp}$) is extracted using an adaptive convolution kernel [6]:

$$\nabla J^{\perp} = -\sum_{a=-1}^{1}\sum_{b=-1}^{1} w_{(i+a,j+b)} * g_{(a,b)} \quad (1)$$

Weights ($w_{(i+a,j+b)}$) based on proximity to the cloud mask are assigned to target image pixels. Cloud-covered pixels are treated as unknowns in a linear system derived from the isophote constraints. Solving this system yields the filled pixel values.

B. *Robust Classification Mechanism*

This study proposes a land cover classification framework for high-resolution remote sensing imagery, which adapts to various classifiers and suppresses class imbalance through a synergistic mechanism of multi-modal feature joint representation and stratified intra-class resampling. The method includes the following core modules:

1) Multi-modal Feature Extraction

This study designs a multi-modal feature joint representation framework, inheriting the core idea of multi-source feature fusion from [7] to enhance robustness through integrating spectral, textural, and statistical features. The feature construction is as follows:

- Color Feature F_{color}: Contains 9 feature dimensions from RGB, HSV, and LAB, directly reflecting the spectral reflectance properties of pixels.
- Textural Feature $F_{texture}$: Based on a sliding window Gray-Level Co-occurrence Matrix (GLCM), it calculates co-occurrence matrices in four directions. The gray level quantization is 8 levels. Statistical indicators include Contrast, Homogeneity, Correlation, and Energy. The features from all directions are averaged and mapped to all pixels within the window. This results in a 4-dimensional feature that represents the local texture structure and replaces Kang's Gabor filtering (48 dimensions) with a compact structural similarity quantifier.
- Multiscale Statistical Features F_{stats}: Based on a pyramid structure of hierarchical regional statistics, it includes 5 types of RGB statistics (mean, standard deviation, skewness, kurtosis, range) and 2 types of grayscale statistics (mean, standard deviation).

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A 20-dimensional joint feature vector is formed through cascade concatenation:

$$F = F_{color} \oplus F_{texture} \oplus F_{texture} \quad (2)$$

2) Bootstrap & Model Training

To address the risk of overfitting in a high-dimensional feature space, a stratified intra-class Bootstrap with replacement ensemble learning framework is proposed to ensure that all classes of samples are fully sampled:

- Training Set Construction: The original class distribution is maintained at an 80% ratio to ensure each class of samples is balanced. The training set is resampled with replacement (sampling rate 80%) to generate 10 differentiated training subsets.
- Base Classifier Design: A random forest is used as the base model, balancing non-linear fitting ability with noise resistance.
- Parallelized Training.
- Ensemble Decision: The class probability outputs of each model are averaged:

$$P(c|X) = \frac{1}{M}\sum_{m=1}^{M} P_m(c|X) \quad (3)$$

- Where M is the total number of models and c is the class label. The class with the maximum probability is selected as the final prediction result:

$$\hat{y} = \arg max_c P(c|X) \quad (4)$$

III. Experiment

The experiments utilized simulated Landsat 8 time-series imagery (5 registered 512×512 images per group, RGB bands) covering diverse topographies. All processing employed standardized radiometric/geometric corrections across Python 3.9 and MATLAB R2023a environments. Three experimental components were designed: 1) land cover classification of original versus reconstructed images, 2) land cover type comparison, and 3) classifier performance analysis - collectively validating algorithm stability and robustness. Quantitative evaluation comprised PSNR and SSIM for assessing cloud-removed imagery, while classification accuracy was measured via difference ratio .

A. Classification Comparison of Original vs. Reconstructed Images

TABLE I. Difference Ratio

	Picture1	Picture2	Picture3	Pictur4	Picture5
Farmland	1.62	3.05	2.04	1.35	1.95
Grassland	0.01	0.28	0.87	0.22	0.01

The binary difference map in figure 4 reveals localized discrepancies in cloud-simulated regions, with farmland (complex terrain) exhibiting notably higher difference ratios than grassland (smooth terrain) in Table I. This performance gap correlates with topographic intricacy, as visually confirmed in figure 4 showing farmland reconstruction.

TABLE II. Cloud Removal Results

Topography	Cloud Amount	Mean SSIM	PSNR
Farmland	6.31%	0.9834	36.0479
	13.82%	0.9695	33.2715
	23.73	0.9457	30.9624
Grasslands	6.31%	0.9919	37.5912
	13.82%	0.9754	33.3068
	23.73	0.9607	32.3576

Quantitative metrics in Table II further demonstrate lower PSNR/SSIM values for farmland across all cloud-coverage levels. Despite these variations, reconstructed textures maintain sharpness with smooth edge transitions to adjacent pixels, preserving spatial continuity in filled regions.

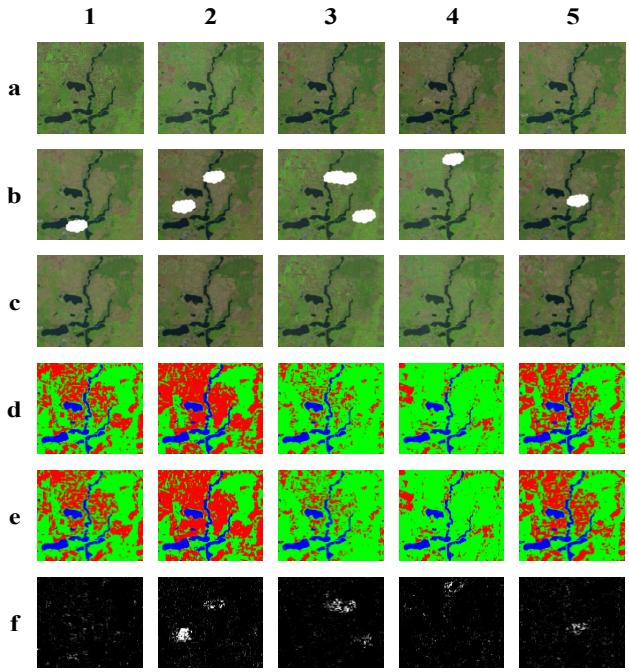


Fig. 4. Experimental result images: (a) original images; (b) simulated cloud images; (c) cloud removal result images; (d) classification of cloud removal results; (e)classification of original images; (f) black and white comparison map of classification results.

B. Multiple Land Cover Types

Figure 5 presents the classification results of different landforms. Comparative experiments on typical terrains (water, grassland, desert) used time-series imagery spanning inter-annual to monthly scales, revealing significant grassland coverage changes. The second image group exhibited blurred land-grassland boundaries with color mixing, causing land misclassification as water. Random forests' sensitivity to abrupt feature changes amplified decision boundaries at texture block edges, generating visible blocking artifacts in results. Reducing texture block size alleviates artifacts but multiplicatively increases computation.

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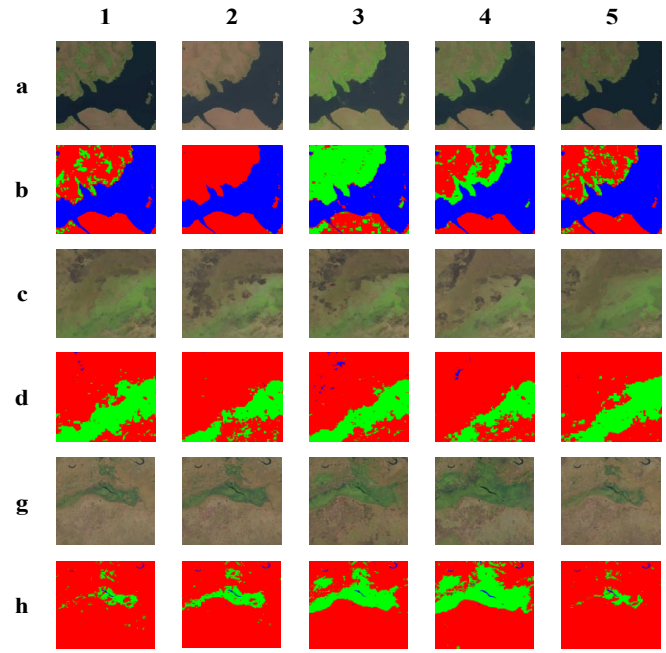


Fig. 5. (a)-(b)Water bodies dominated,(c)-(d)Grasslands,(g)-(h)deserts.

C. Comparison of Multiple Classifiers

TABLE III. DIFFERENCE RATIO OF CLASSIFIERS

	Picture1	Picture2	Picture3	Pictur4	Picture5
Svm	5.29	5.38	4.01	3.88	4.94
Knn	5.58	7.12	5.20	4.60	7.55

In this experiment, considering the characteristics of SVM and KNN classifiers, the feature values were dimensionally reduced by discarding texture features. In terms of time complexity, KNN was the fastest, while SVM was much slower than the other two classifiers. As shown in Table III, the classification results of SVM, KNN, and RF show no significant differences, demonstrating the universality and robustness of the proposed method. As shown in Figure 6&7, among them, KNN's classification results have a large number of scattered points, while SVM's are the most regular, and RF is in between.

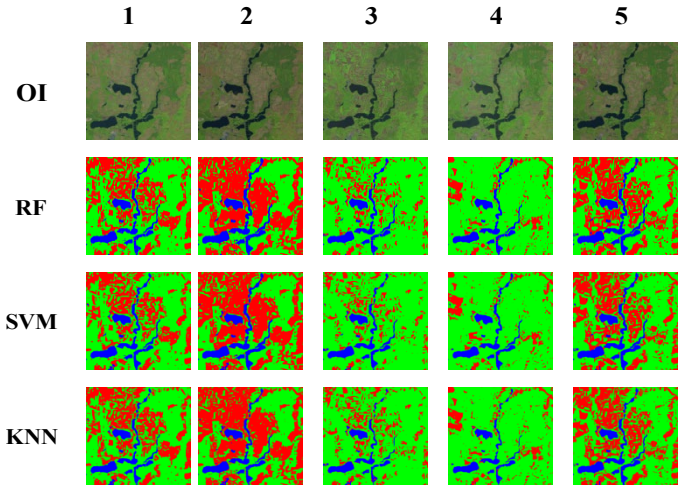


Fig. 6. Comparison chart of different classifiers.OI means “original images”.

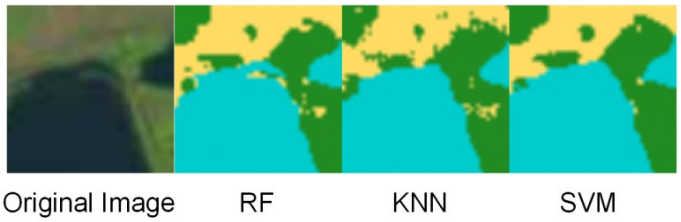


Fig. 7. Comparison of the performance of different classifiers for local details. As seen in the figure, under the same configuration, RF can use more feature values, and its classification results have more detail than the other two classifiers.

IV. CONCLUSION

To address cloud-contaminated land cover classification in time-series remote sensing images, this paper proposes the dual-path cloud removal and multi-modal classification method, featuring key contributions: an cloud removal mechanism that analyzes cloud mask temporal relationships via logical & operations to determine repairability and optimizes reference image selection with a middle-frame priority principle; a combined approach utilizing superpixel-based regression for cloud-overlap areas and isophote networks to ensure spatial continuity in non-overlapping regions; and enhanced feature representation through multi-scale joint features for spectral-texture enhancement, employing stratified Bootstrap sampling to generate balanced training subsets combined with random forest ensemble decision-making to alleviate class imbalance and overfitting.

However, the method used in this paper only uses the RGB three channels for classification, making it difficult to handle land cover conditions with blurred boundaries and mixed colors, such as water bodies appearing in different areas as blue to black, or even green. In such cases, classifying based on color alone can easily lead to misclassification as other types. Future work could continue to add more channels to achieve more accurate classification.

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